تحلیل رابطه بین تحصیلات سرپرست خانوار و رفاه اقتصادی خانوارهای ایرانی



گروه مردد در تردد

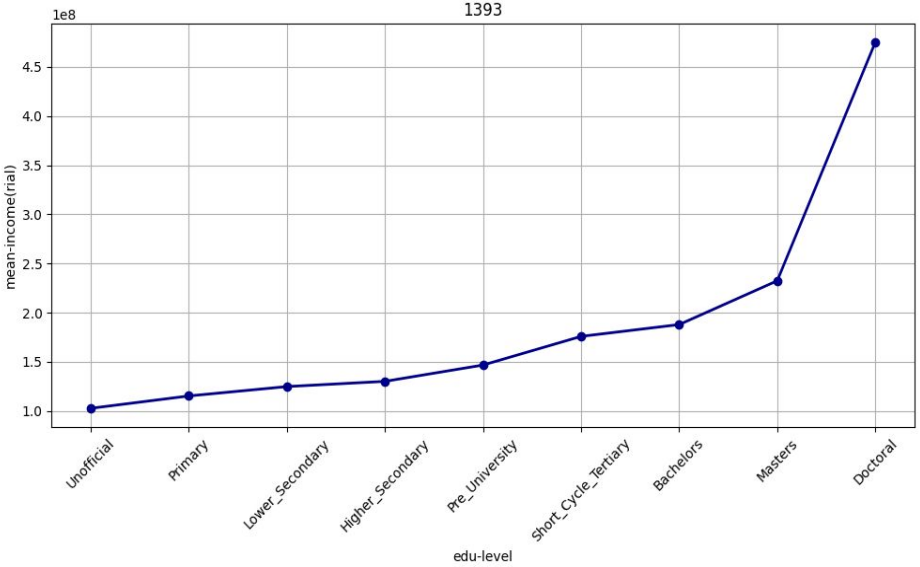
- علی عبدلی مسینان
- محمدرضا رحمانی
- پارسا محمدی تربتی

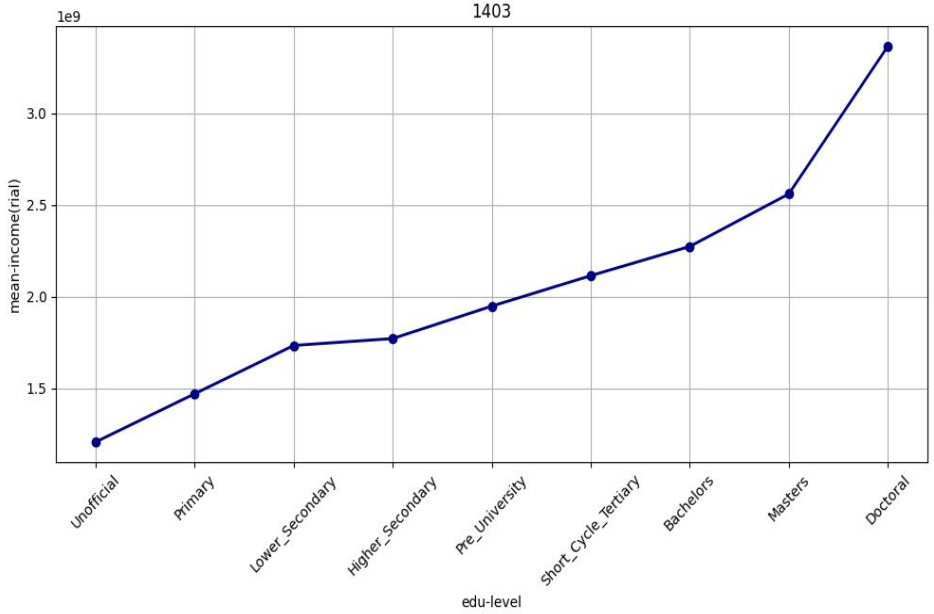
منابع داده

داده‌های هزینه و درآمد خانوار(HBS)

- جداول Members_Total_Income و members_properties و Expenditures و employment_income
- سال‌های ۱۳۹۳ تا ۱۴۰۳

با هم ببینیم


1393
1e8
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
mean-income(rial)
Unofficial
Primary
Lower_Secondary
Higher_Secondary
Pre_University
Short_Cycle_Tertiary
Bachelors
Masters
Doctoral
edu-level

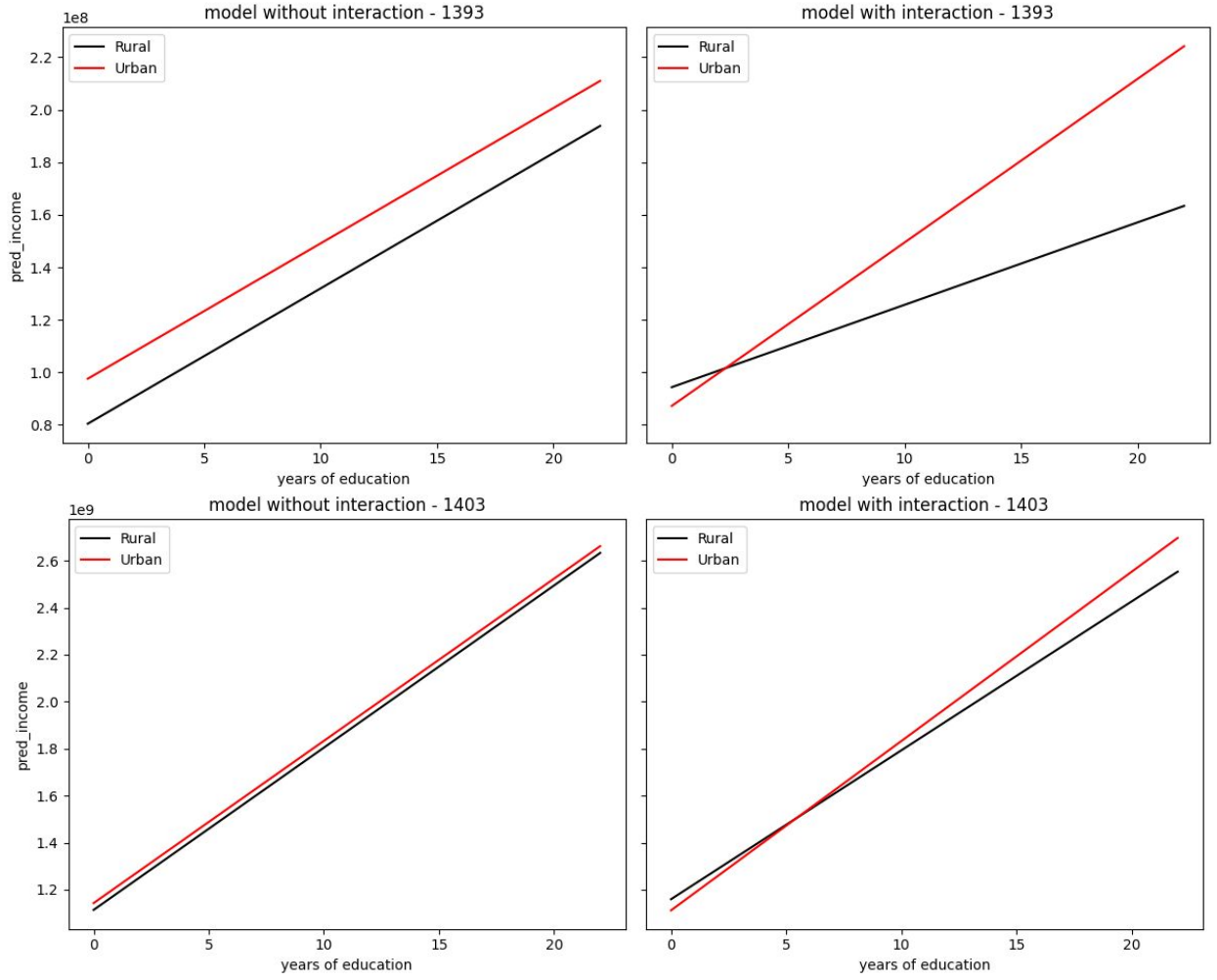

1403
1e9
3.0
2.5
2.0
1.5
mean-income(rial)
Unofficial
Primary
Lower_Secondary
Higher_Secondary
Pre_University
Short_Cycle_Tertiary
Bachelors
Masters
Doctoral
edu-level

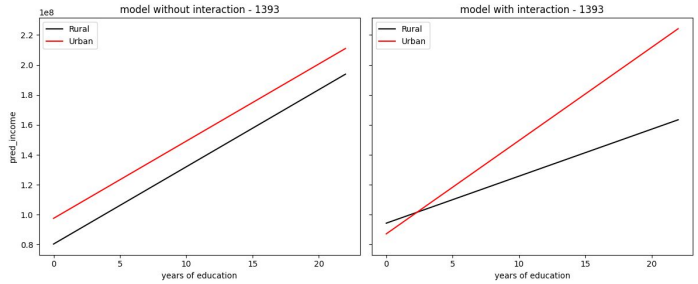
مدل بدون تعامل:

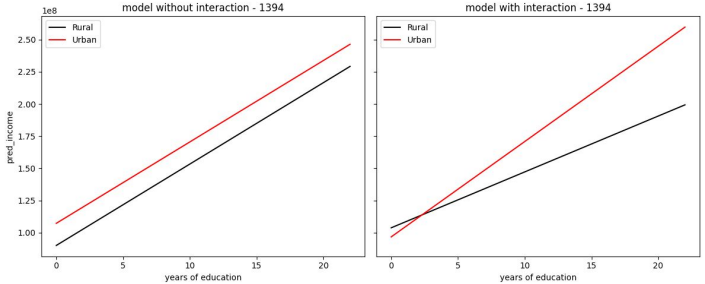
$$Y = \beta_0 + \beta_1 \cdot Edu + \beta_2 \cdot X + \varepsilon$$

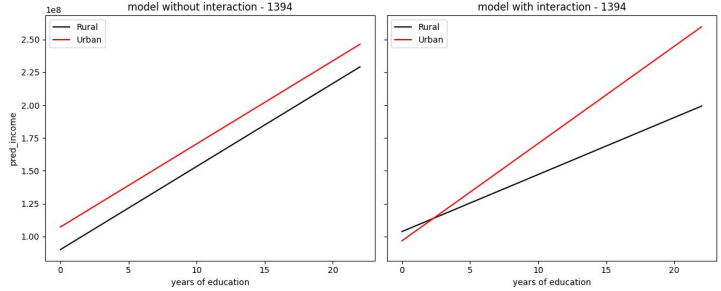
مدل با تعامل:

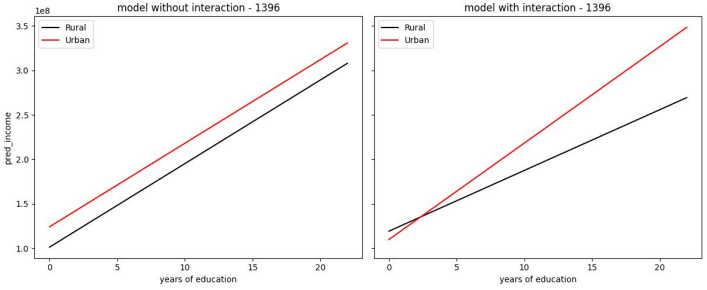
$$Y = \beta_0 + \beta_1 \cdot Edu + \beta_2 \cdot X + \beta_3 \cdot (Edu \times X) + \varepsilon$$

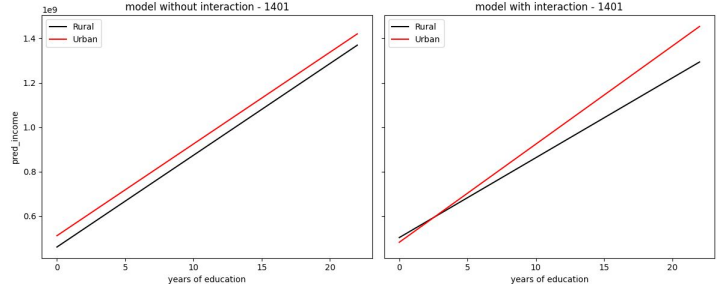

model without interaction - 1393
1e8
Rural
Urban
pred_income
years of education
model with interaction - 1393
Rural
Urban
years of education
model without interaction - 1403
1e9
Rural
Urban
pred_income
years of education
model with interaction - 1403
Rural
Urban
years of education

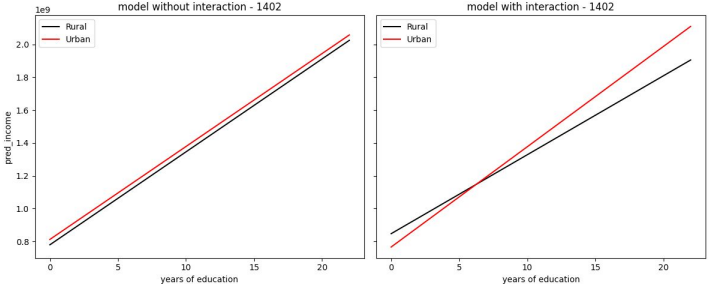

model without interaction - 1393
model with interaction - 1393
1e8
Rural
Urban
pred_income
years of education

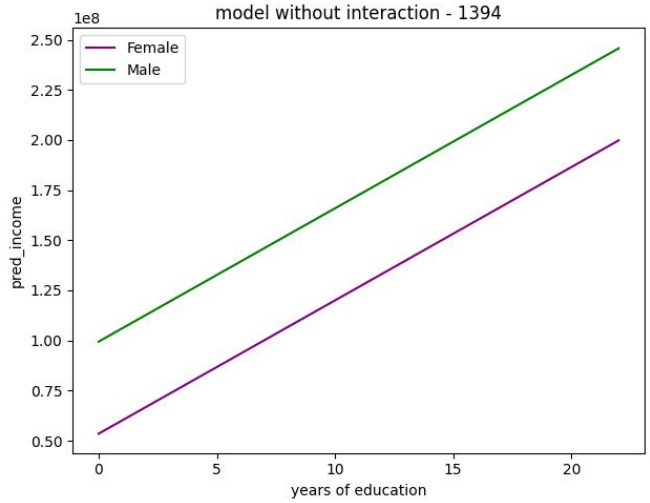

model without interaction - 1394
model with interaction - 1394
1e8
Rural
Urban
pred_income
years of education

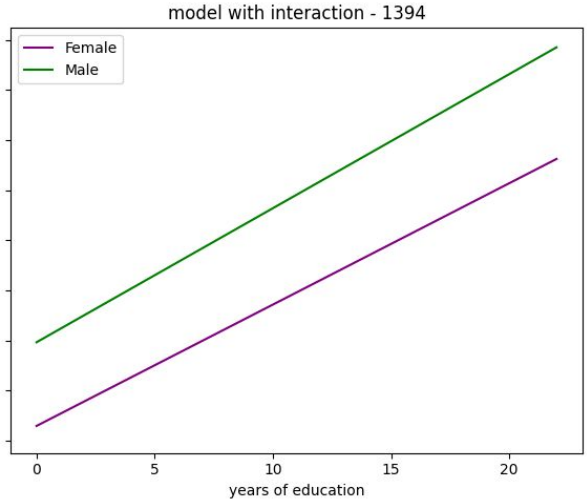

model without interaction - 1394
model with interaction - 1394
1e8
Rural
Urban
pred_income
years of education

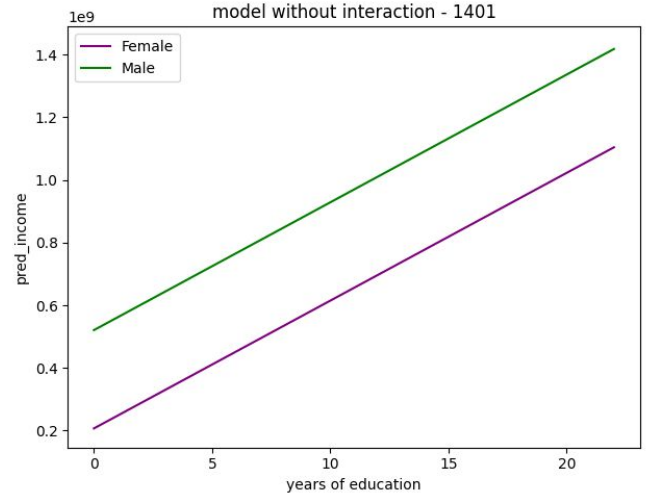

model without interaction - 1396
model with interaction - 1396
1e8
Rural
Urban
pred_income
years of education

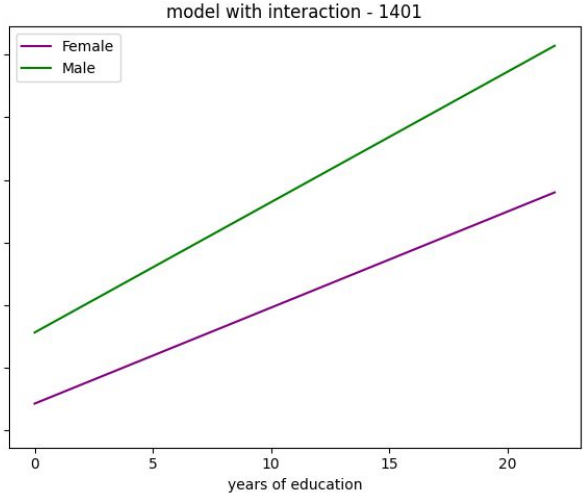

model without interaction - 1401
model with interaction - 1401
1e9
Rural
Urban
pred_income
years of education

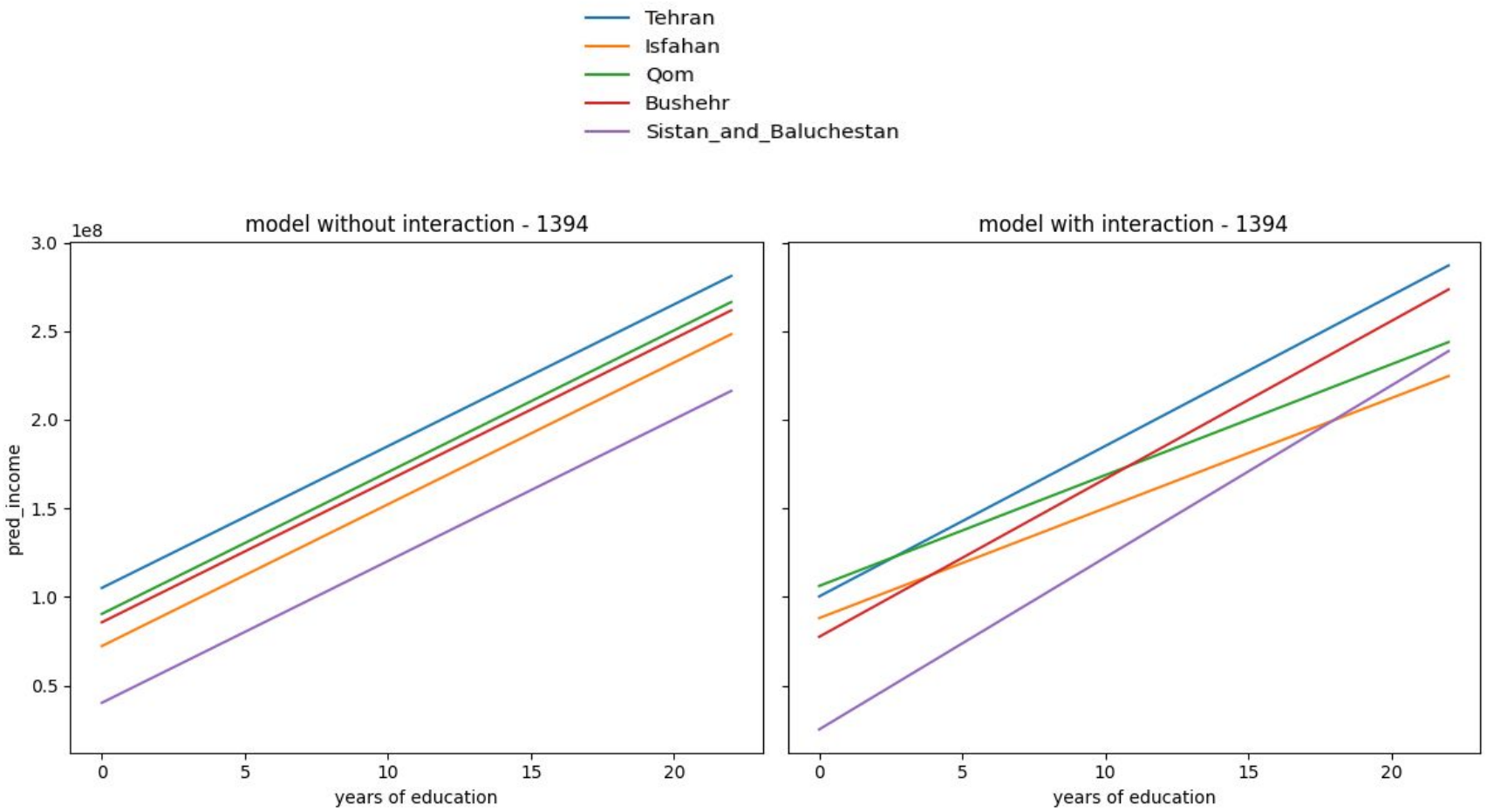

model without interaction - 1402
model with interaction - 1402
1e9
Rural
Urban
pred_income
years of education

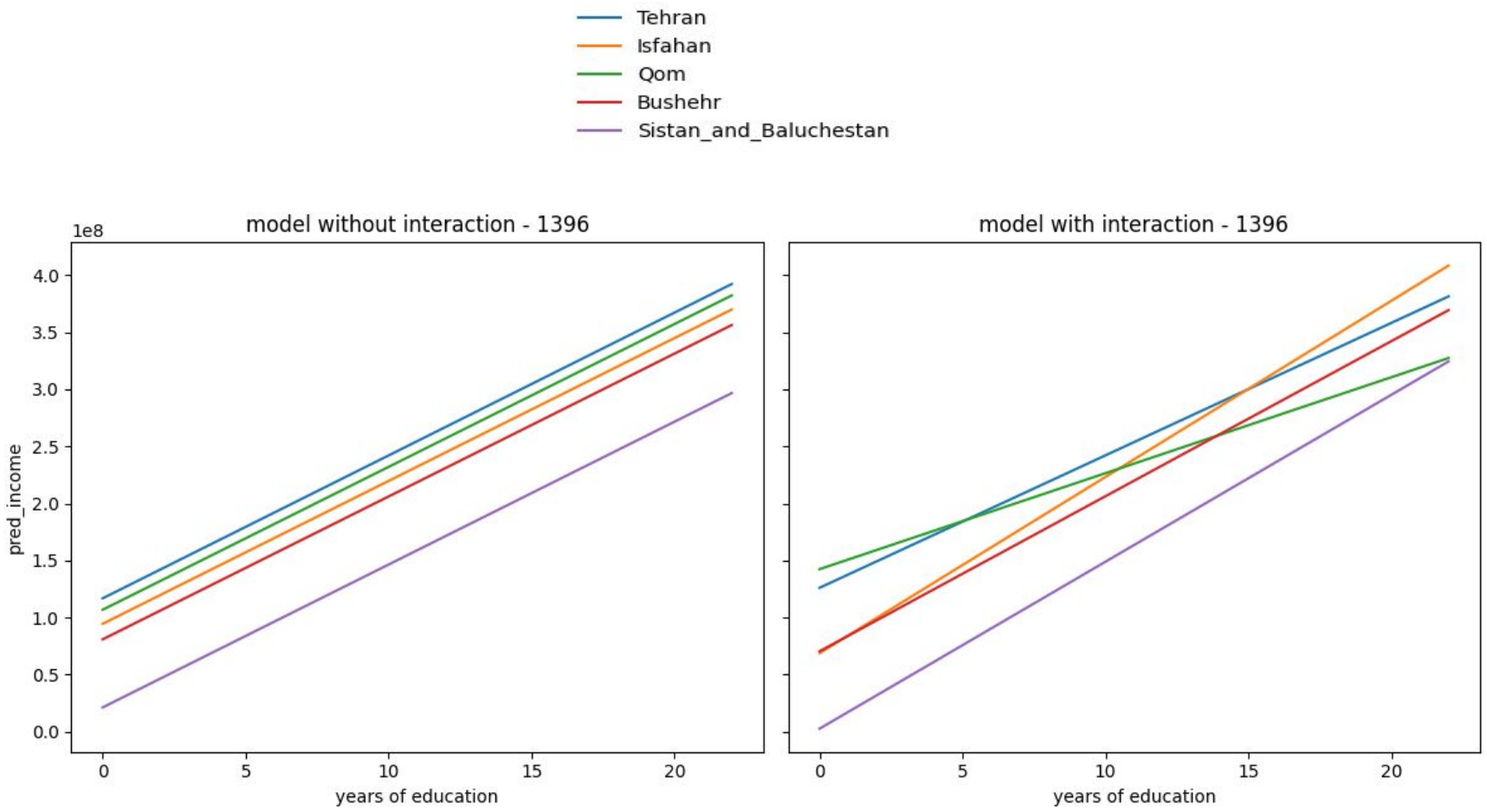

model without interaction - 1394
1e8
Female
Male
pred_income
years of education

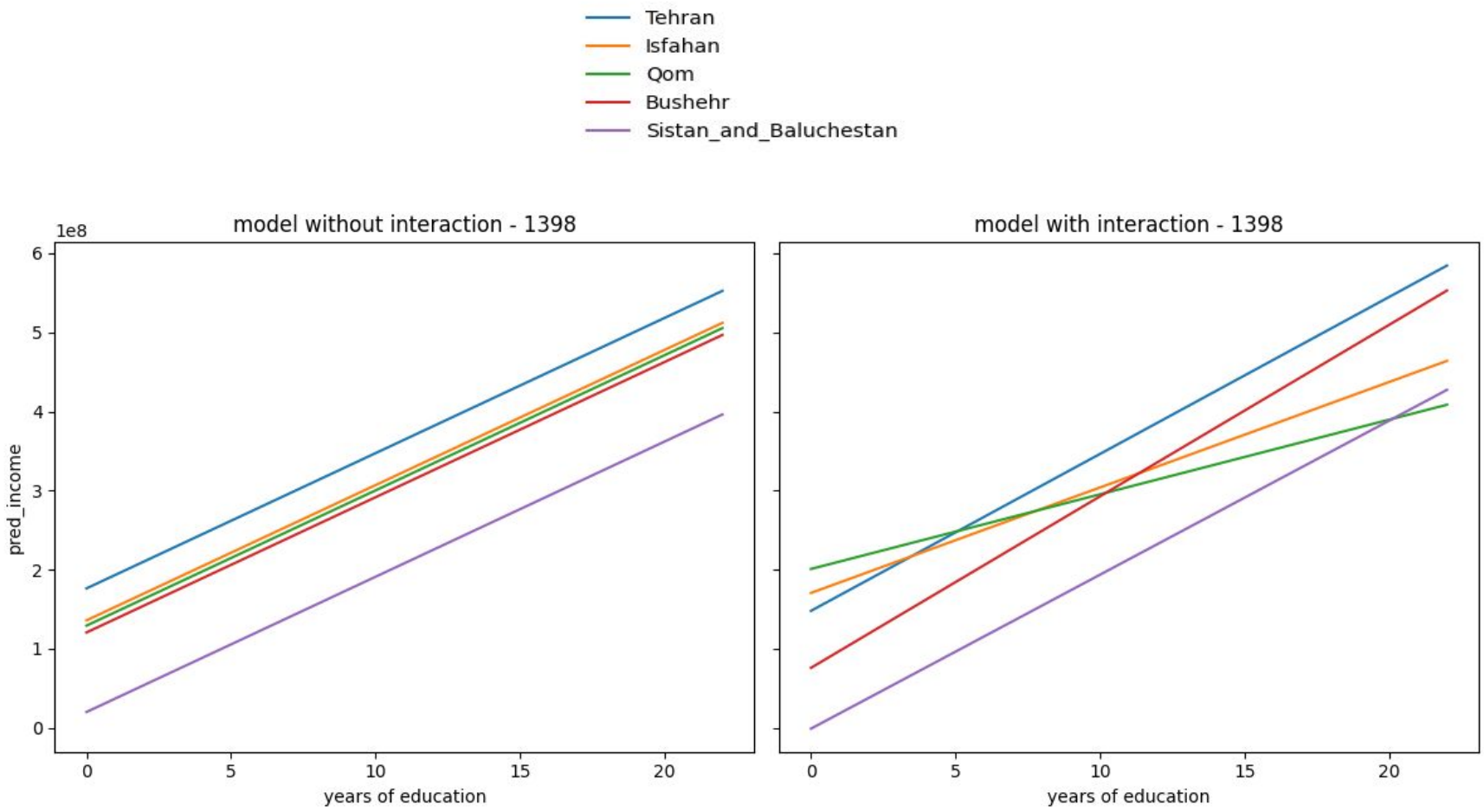

model with interaction - 1394
Female
Male
years of education

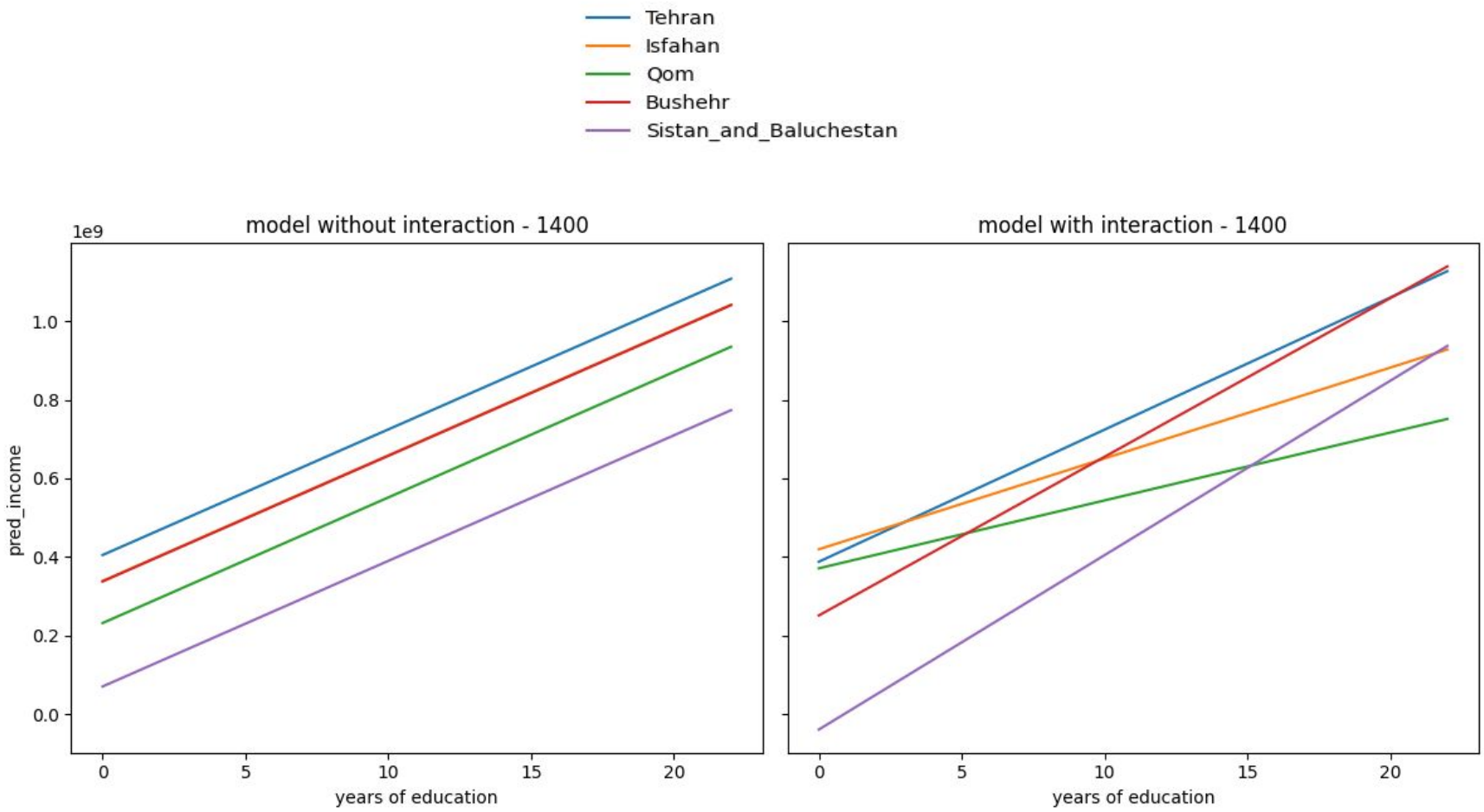

model without interaction - 1401
1e9
Female
Male
pred_income
years of education

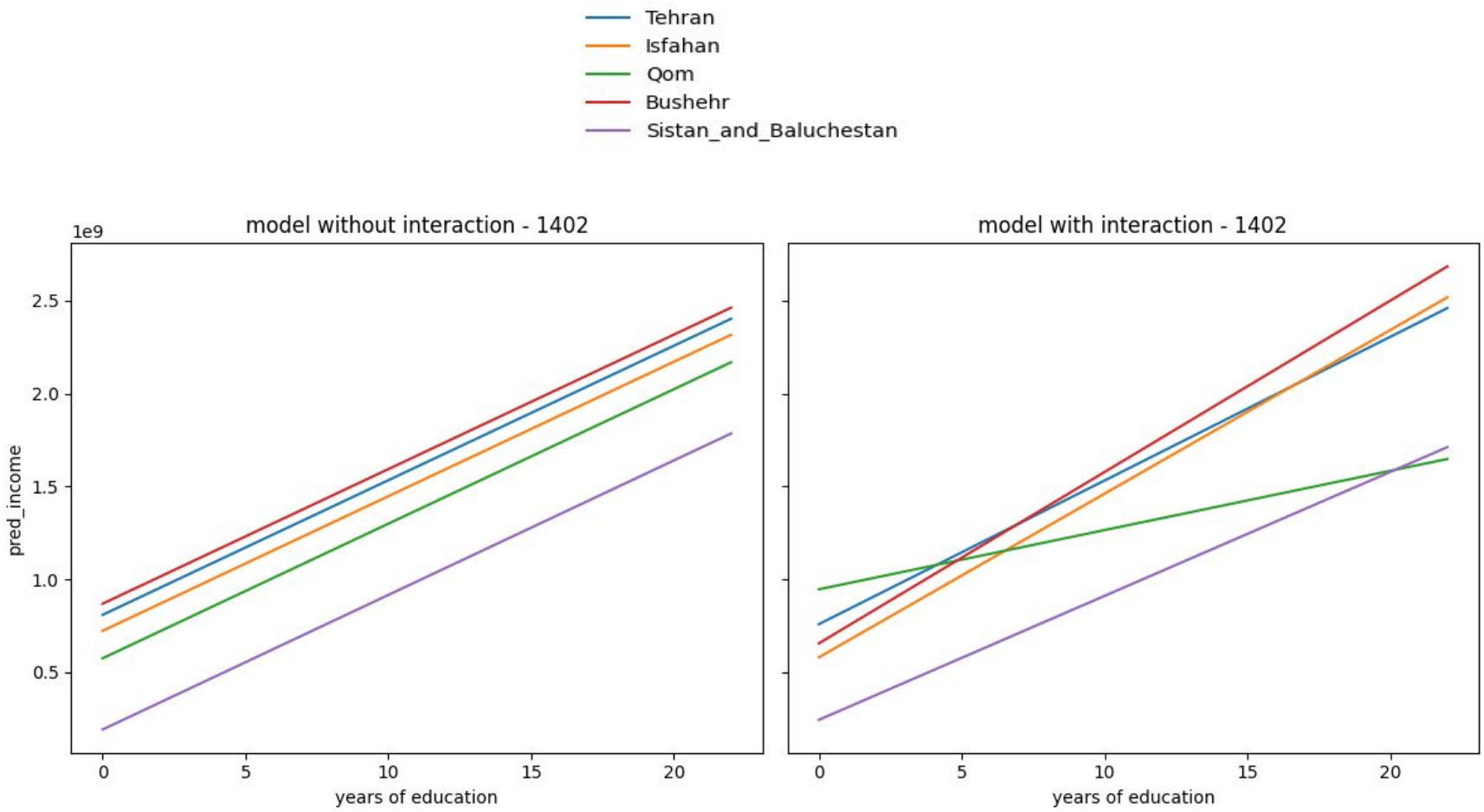

model with interaction - 1401
Female
Male
years of education

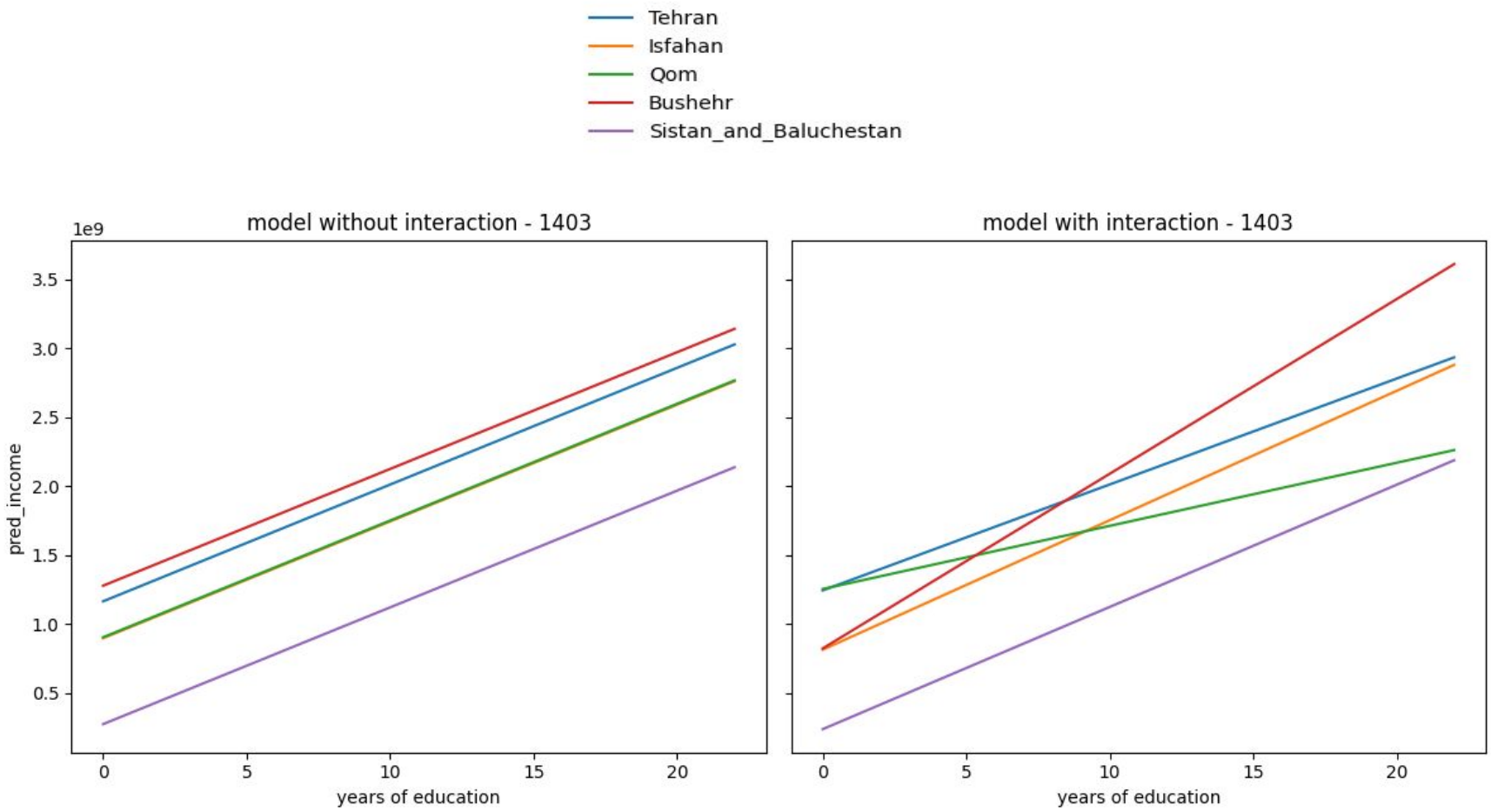

Tehran
Isfahan
Qom
Bushehr
Sistan_and_Baluchestan
model without interaction - 1394
model with interaction - 1394
1e8
3.0
2.5
2.0
1.5
1.0
0.5
pred_income
0
5
10
15
20
years of education
years of education


Tehran
Isfahan
Qom
Bushehr
Sistan_and_Baluchestan
model without interaction - 1396
model with interaction - 1396
1e8
pred_income
years of education
years of education


Tehran
Isfahan
Qom
Bushehr
Sistan_and_Baluchestan
model without interaction - 1398
model with interaction - 1398
1e8
pred_income
years of education
years of education


Tehran
Isfahan
Qom
Bushehr
Sistan_and_Baluchestan
model without interaction - 1400
model with interaction - 1400
1e9
pred_income
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
years of education
years of education


Tehran
Isfahan
Qom
Bushehr
Sistan_and_Baluchestan
model without interaction - 1402
model with interaction - 1402
1e9
pred_income
2.5
2.0
1.5
1.0
0.5
0
5
10
15
20
years of education
years of education


Tehran
Isfahan
Qom
Bushehr
Sistan_and_Baluchestan
model without interaction - 1403
model with interaction - 1403
1e9
pred_income
years of education
years of education

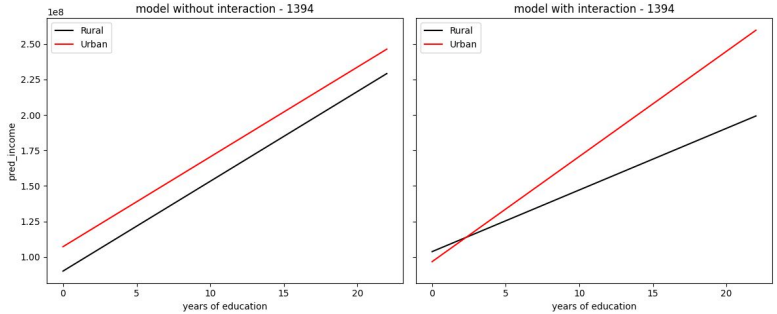
شاخص رفاه (PPI)

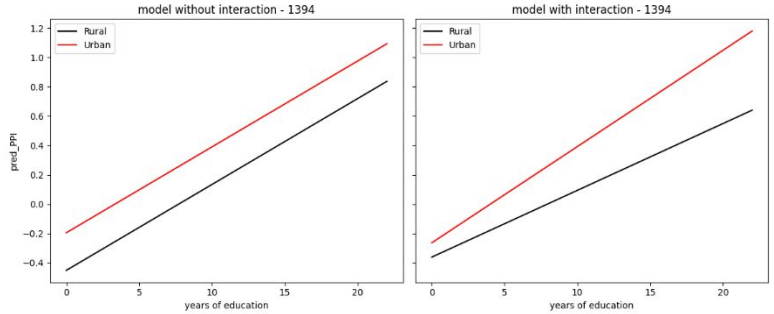
شاخص رفاه (PPI)

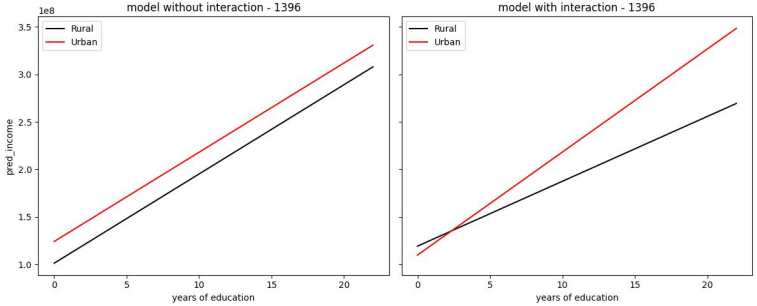
چرا؟

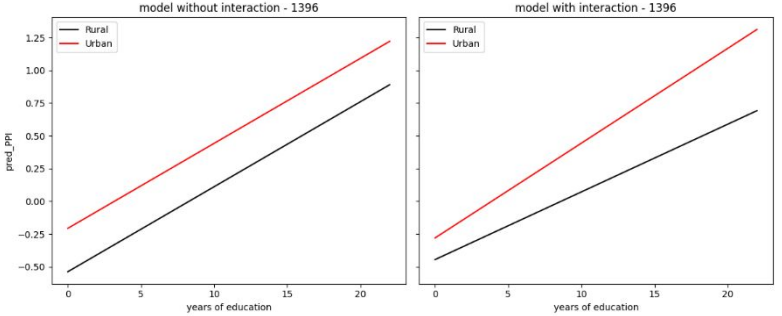
ساخت شاخص رفاه

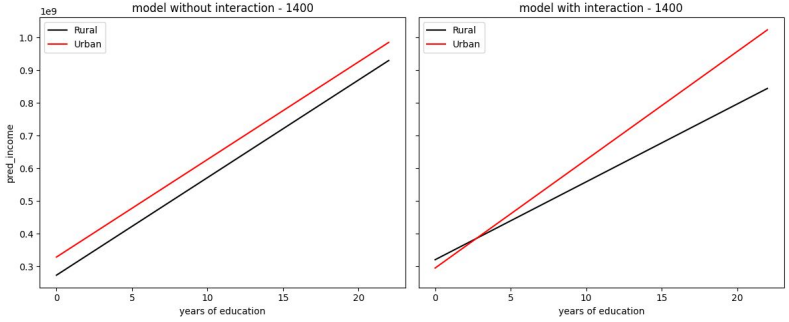
- درآمد واقعی سرانه
  - شاخص قیمت منطقه‌ای
  - اندازه معادل خانوار
- هزینه واقعی سرانه
- سهم خوراک
- سهم مسکن

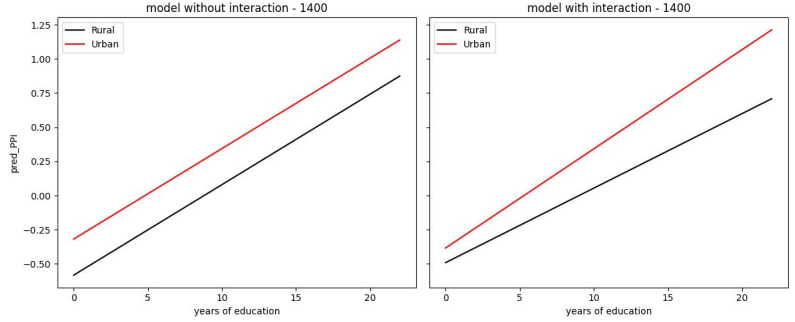

model without interaction - 1394
model with interaction - 1394
Rural
Urban
1e8
pred_income
years of education

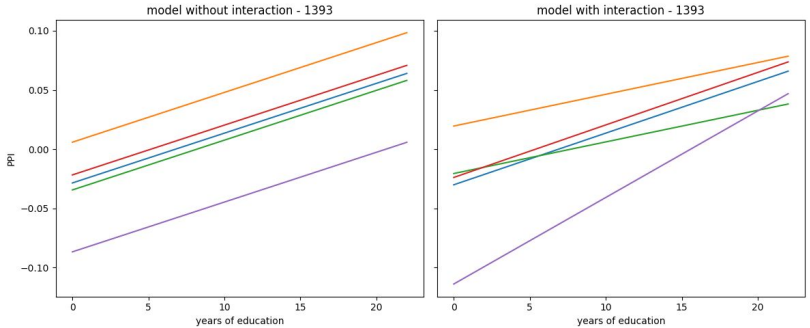

model without interaction - 1394
model with interaction - 1394
Rural
Urban
pred_PPI
years of education


model without interaction - 1396
model with interaction - 1396
Rural
Urban
1e8
pred_income
years of education


model without interaction - 1396
model with interaction - 1396
Rural
Urban
pred_PPI
years of education

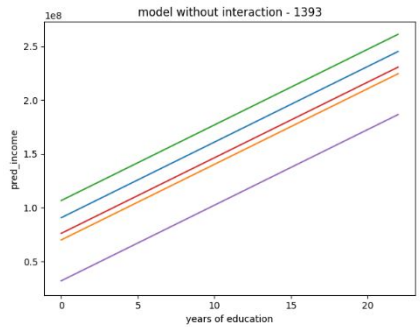

model without interaction - 1400
model with interaction - 1400
Rural
Urban
1e9
pred_income
years of education

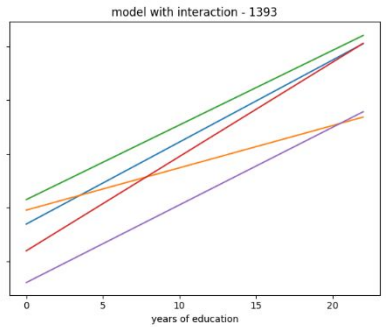

model without interaction - 1400
model with interaction - 1400
Rural
Urban
pred_PPI
years of education

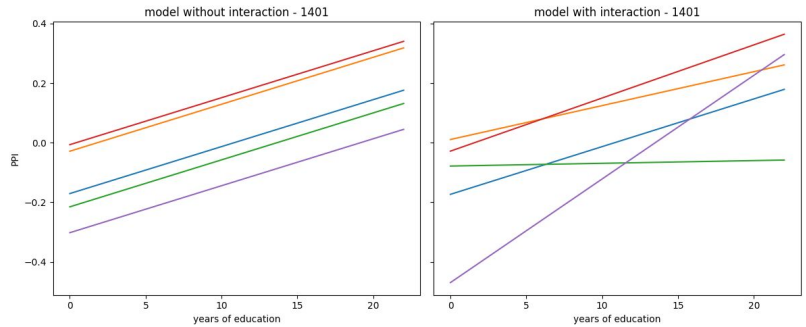

model without interaction - 1393
PPI
years of education

model with interaction - 1393
years of education

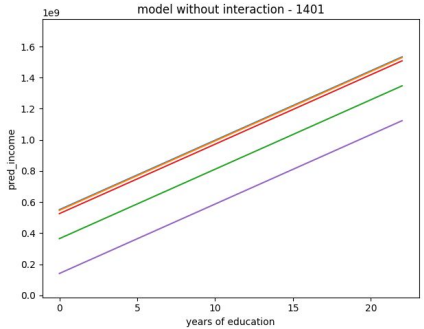

Tehran
Isfahan
Qom
Bushehr
Sistan_and_Baluchestan

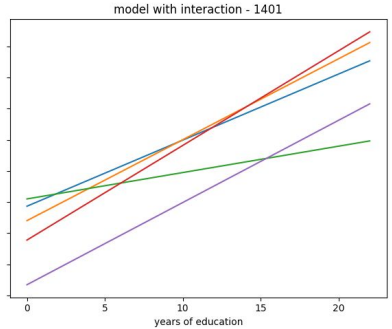

1e8
model without interaction - 1393
pred_income
years of education


model with interaction - 1393
years of education


model without interaction - 1401
PPI
years of education

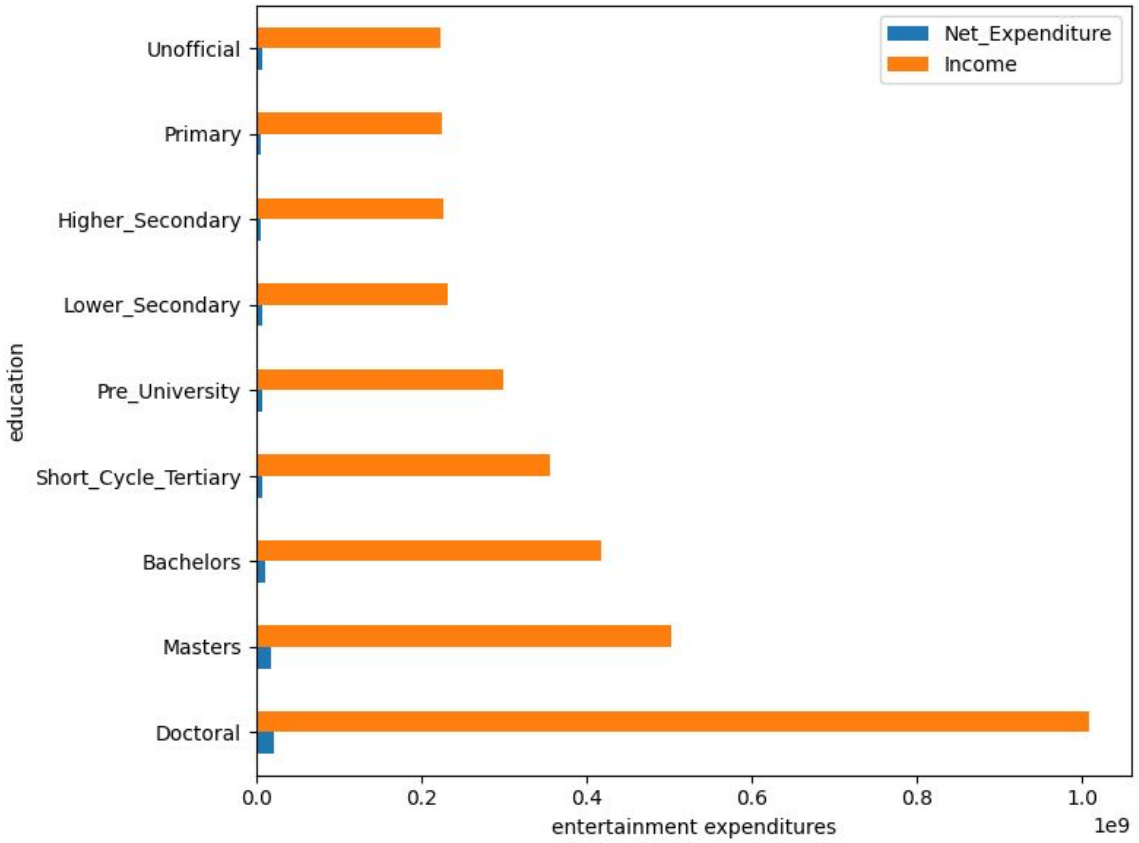
model with interaction - 1401
years of education


1e9
model without interaction - 1401
pred_income
years of education


model with interaction - 1401
years of education

تحلیل تاثیر تحصیلات بر الگوهای هزینه

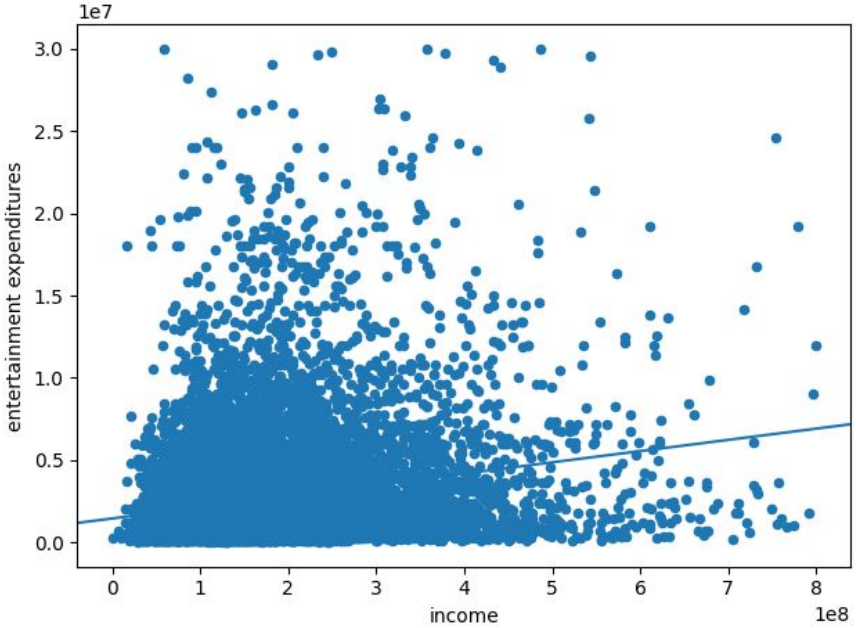
آیا خانواده‌هایی که هزینه‌ی بیشتری برای تفریح می‌کنند، از سطح تحصیلات بالاتری برخوردارند؟


Net_Expenditure
Income
Unofficial
Primary
Higher_Secondary
Lower_Secondary
Pre_University
Short_Cycle_Tertiary
Bachelors
Masters
Doctoral
education
0.0
0.2
0.4
0.6
0.8
1.0
1e9
entertainment expenditures

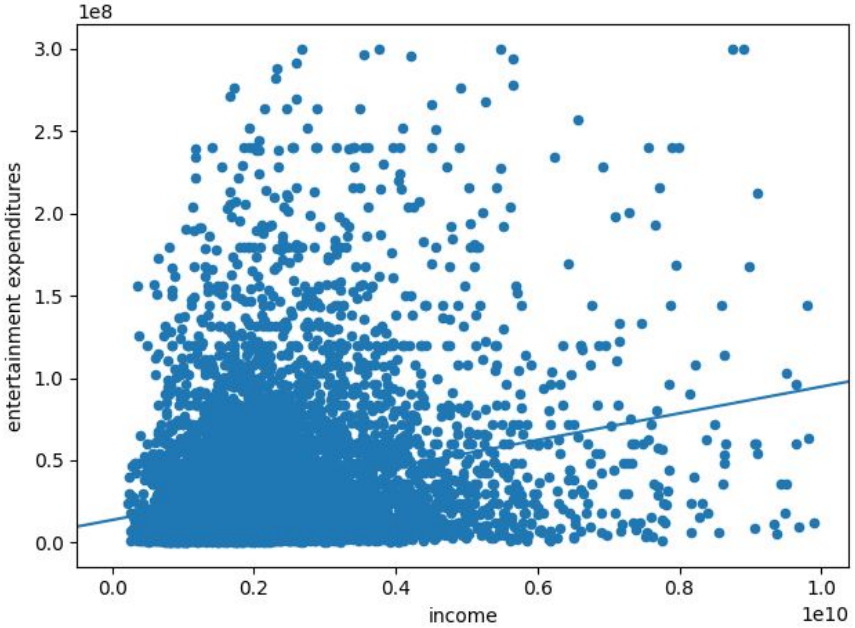
مقایسه تغییر پارامترهای رگرسیون خطی پس از افزودن درآمد به عنوان متغیر پیشگو

	coef	std err	t	P>|t|	[0.025	0.975]
const	4.016e+07	2.81e+06	14.278	0.000	3.46e+07	4.57e+07
Education_Level_Doctoral	9.115e+07	2e+07	4.563	0.000	5.2e+07	1.3e+08
Education_Level_Higher_Secondary	-9.767e+06	8.29e+06	-1.178	0.239	-2.6e+07	6.49e+06
Education_Level_Lower_Secondary	-9.224e+06	3.82e+06	-2.416	0.016	-1.67e+07	-1.74e+06
Education_Level_Masters	2.323e+07	6.72e+06	3.455	0.001	1.01e+07	3.64e+07
Education_Level_Pre_University	-6.041e+06	3.7e+06	-1.631	0.103	-1.33e+07	1.22e+06
Education_Level_Primary	-1.341e+07	3.7e+06	-3.623	0.000	-2.07e+07	-6.15e+06
Education_Level_Short_Cycle_Tertiary	2.717e+07	6.35e+06	4.278	0.000	1.47e+07	3.96e+07
Education_Level_Unofficial	-2.271e+07	3.44e+07	-0.661	0.509	-9.01e+07	4.47e+07

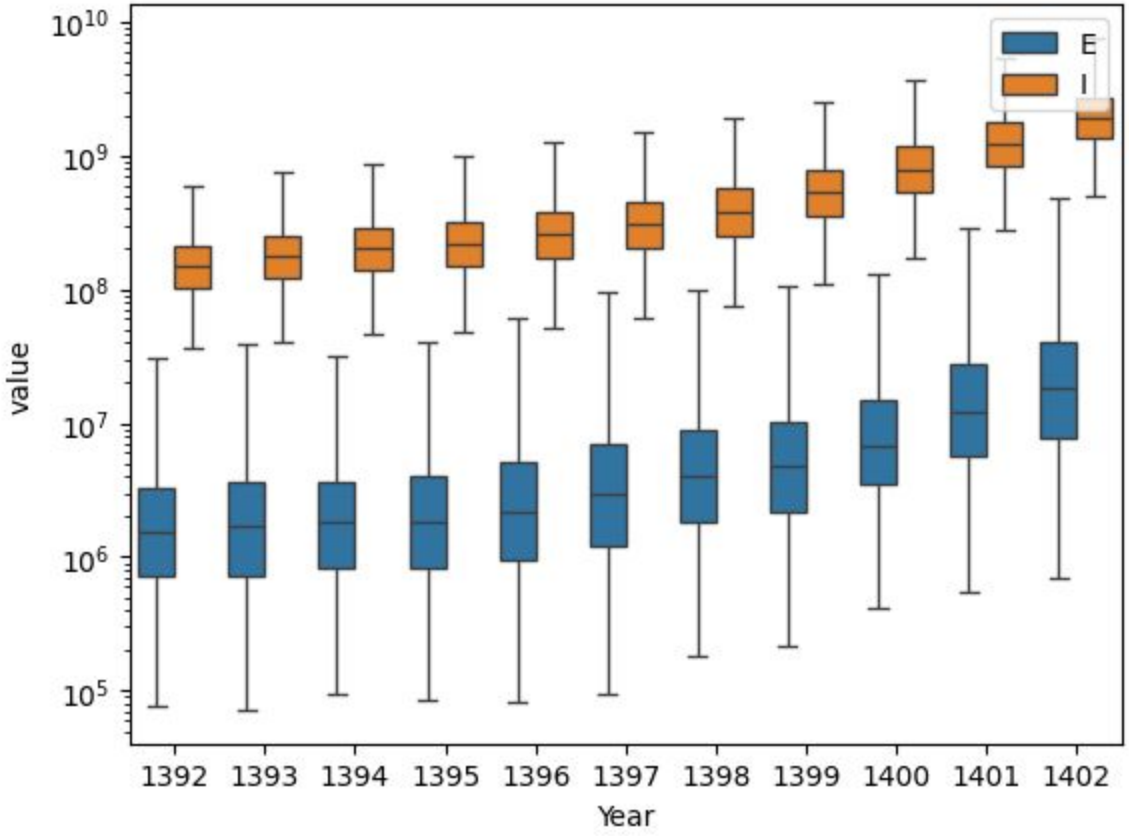
	coef	std err	t	P>|t|	[0.025	0.975]
const	1.308e+07	4.93e+06	2.652	0.008	3.41e+06	2.27e+07
Education_Level_Doctoral	2.803e+07	2.76e+07	1.017	0.309	-2.6e+07	8.21e+07
Education_Level_Higher_Secondary	9.967e+06	1.35e+07	0.740	0.460	-1.65e+07	3.64e+07
Education_Level_Lower_Secondary	-2.343e+06	7.16e+06	-0.327	0.743	-1.64e+07	1.17e+07
Education_Level_Masters	1.322e+07	1.28e+07	1.033	0.302	-1.19e+07	3.83e+07
Education_Level_Pre_University	5.164e+06	6.67e+06	0.774	0.439	-7.91e+06	1.82e+07
Education_Level_Primary	5.827e+06	6.07e+06	0.960	0.337	-6.06e+06	1.77e+07
Education_Level_Short_Cycle_Tertiary	7.041e+05	1.39e+07	0.051	0.960	-2.65e+07	2.79e+07
Education_Level_Unofficial	-8.371e+06	9.66e+07	-0.087	0.931	-1.98e+08	1.81e+08
Income	0.0106	0.002	6.664	0.000	0.007	0.014
Education_Level_Doctoral_&_Income	0.0040	0.003	1.170	0.242	-0.003	0.011
Education_Level_Higher_Secondary_&_Income	-0.0069	0.005	-1.333	0.183	-0.017	0.003
Education_Level_Lower_Secondary_&_Income	-0.0004	0.003	-0.139	0.889	-0.006	0.005
Education_Level_Masters_&_Income	-0.0003	0.003	-0.085	0.932	-0.007	0.006
Education_Level_Pre_University_&_Income	-0.0036	0.002	-1.550	0.121	-0.008	0.001
Education_Level_Primary_&_Income	-0.0065	0.002	-3.137	0.002	-0.011	-0.002
Education_Level_Short_Cycle_Tertiary_&_Income	0.0100	0.005	2.104	0.035	0.001	0.019
Education_Level_Unofficial_&_Income	-0.0040	0.046	-0.087	0.931	-0.095	0.087


1e7
3.0
2.5
2.0
1.5
1.0
0.5
0.0
entertainment expenditures
0
1
2
3
4
5
6
7
8
income
1e8

۱۳۹۲

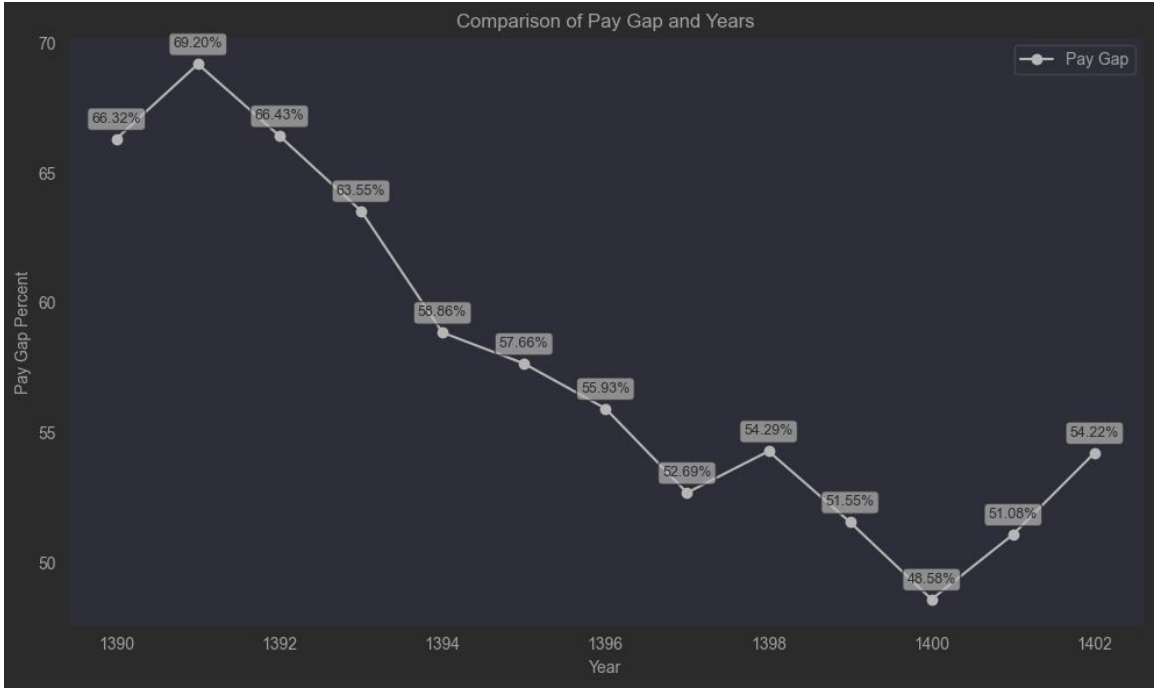

1e8
3.0
2.5
2.0
1.5
1.0
0.5
0.0
entertainment expenditures
0.0
0.2
0.4
0.6
0.8
1.0
income
1e10

۱۴۰۲

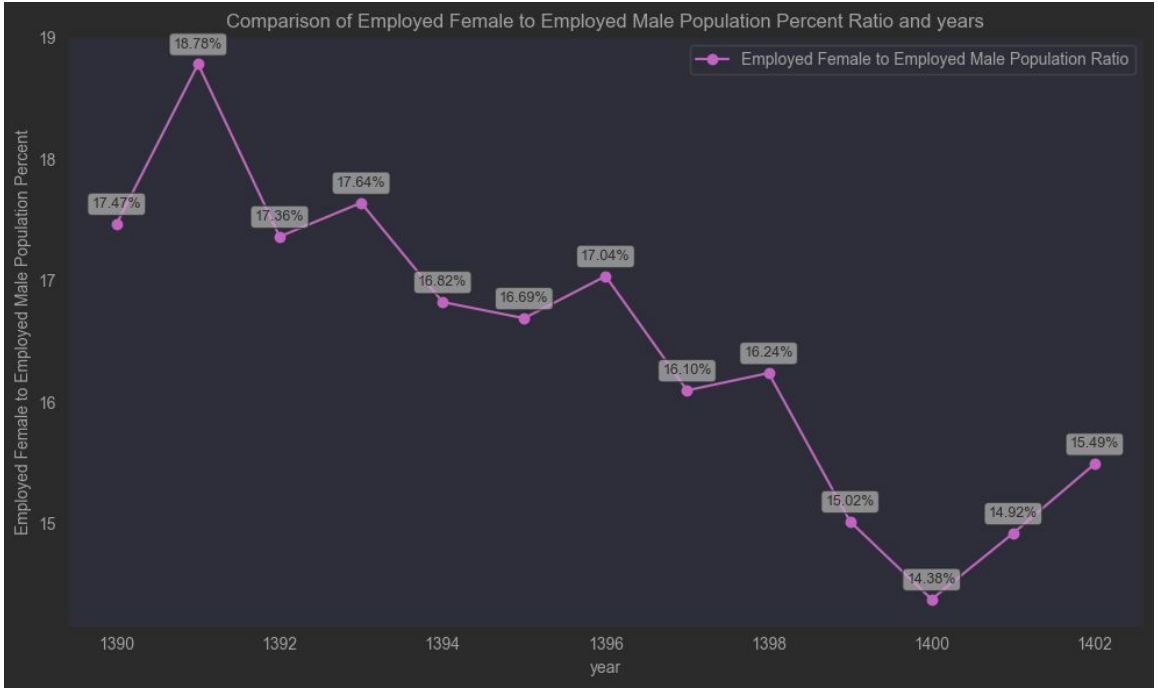

E
I
value
Year
10^{10}
10^{9}
10^{8}
10^{7}
10^{6}
10^{5}
1392
1393
1394
1395
1396
1397
1398
1399
1400
1401
1402

تحلیل اشتغال و درآمد زنان و مردان با استفاده از شاخص های اقتصادی

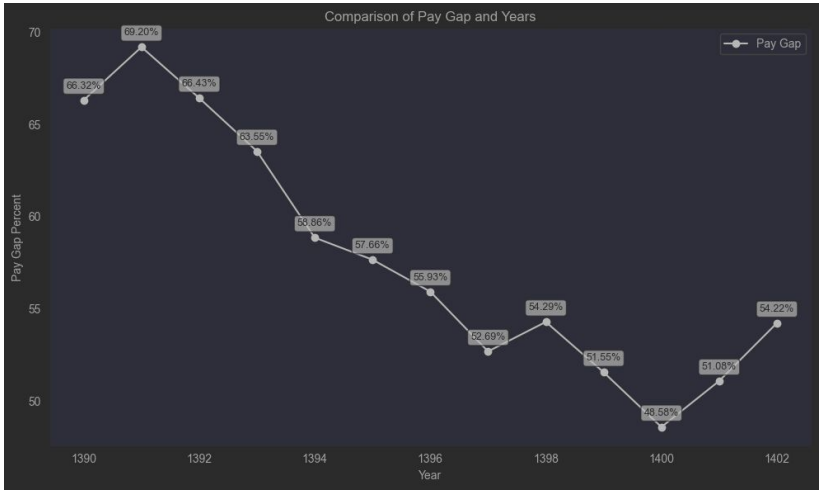
$$\text{شکاف دستمزد} = \left(1 - \frac{\text{میانگین درآمد زنان}}{\text{میانگین درآمد مردان}}\right) \times 100$$

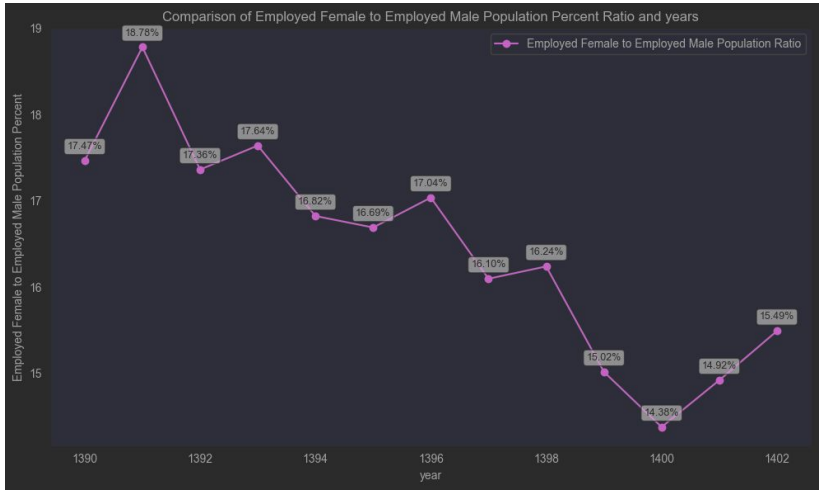


$$\text{نسبت اشتغال} = \frac{\text{تعداد زنان شاغل}}{\text{تعداد مردان شاغل}} \times 100$$

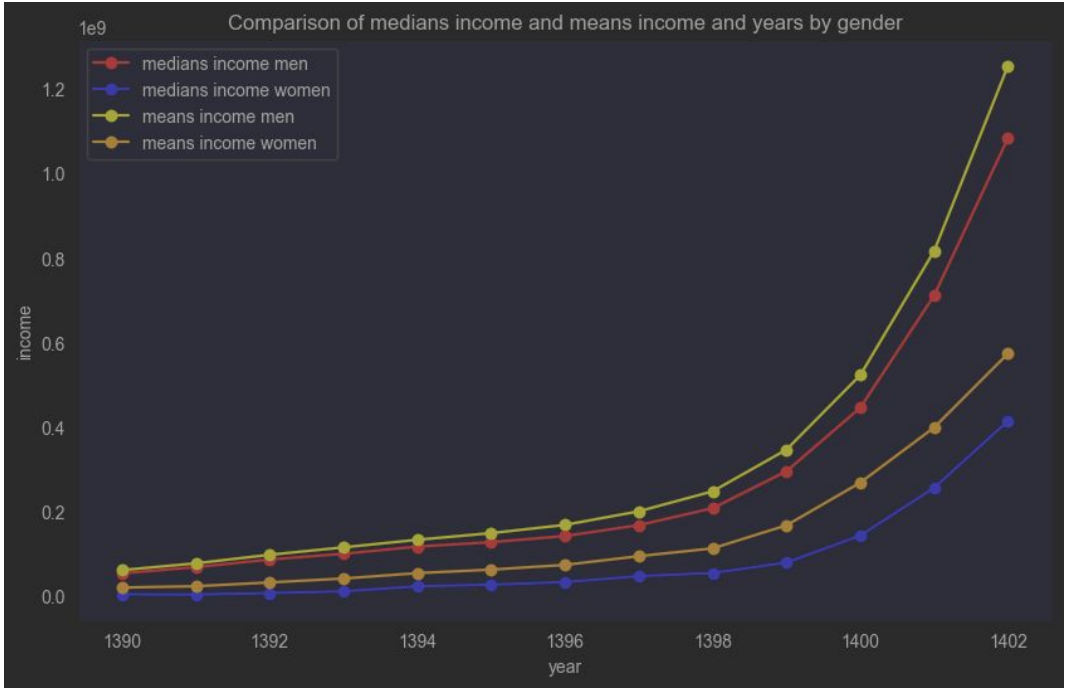


تناقض: پیشرفت یا پسرفت

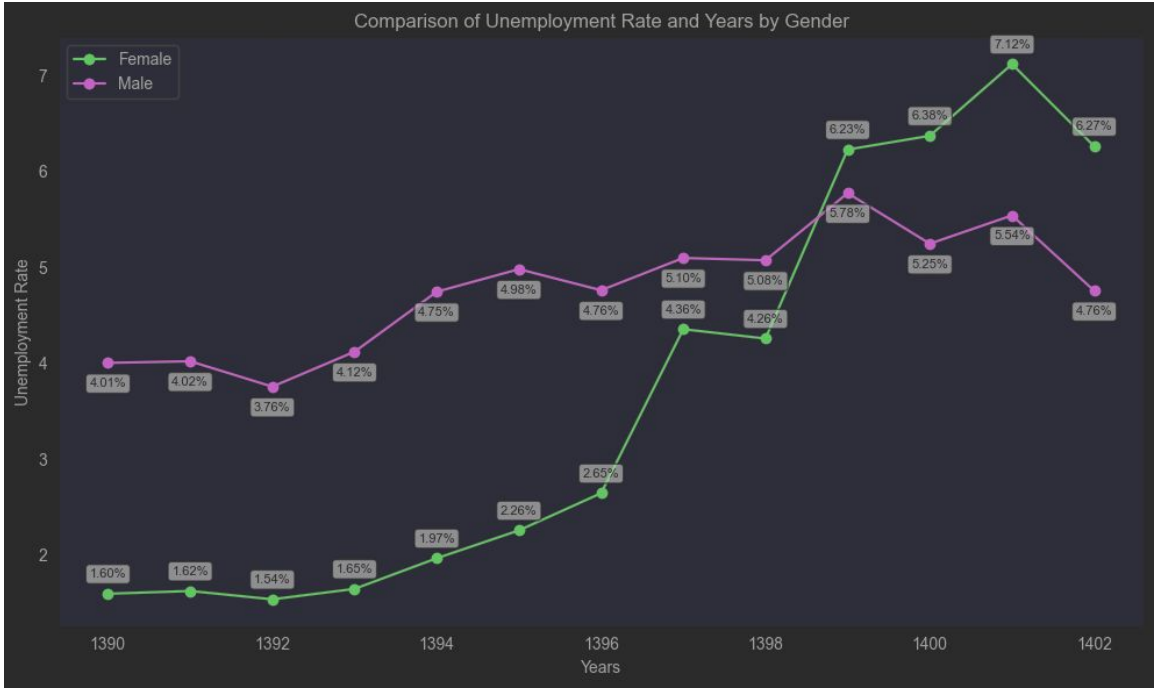




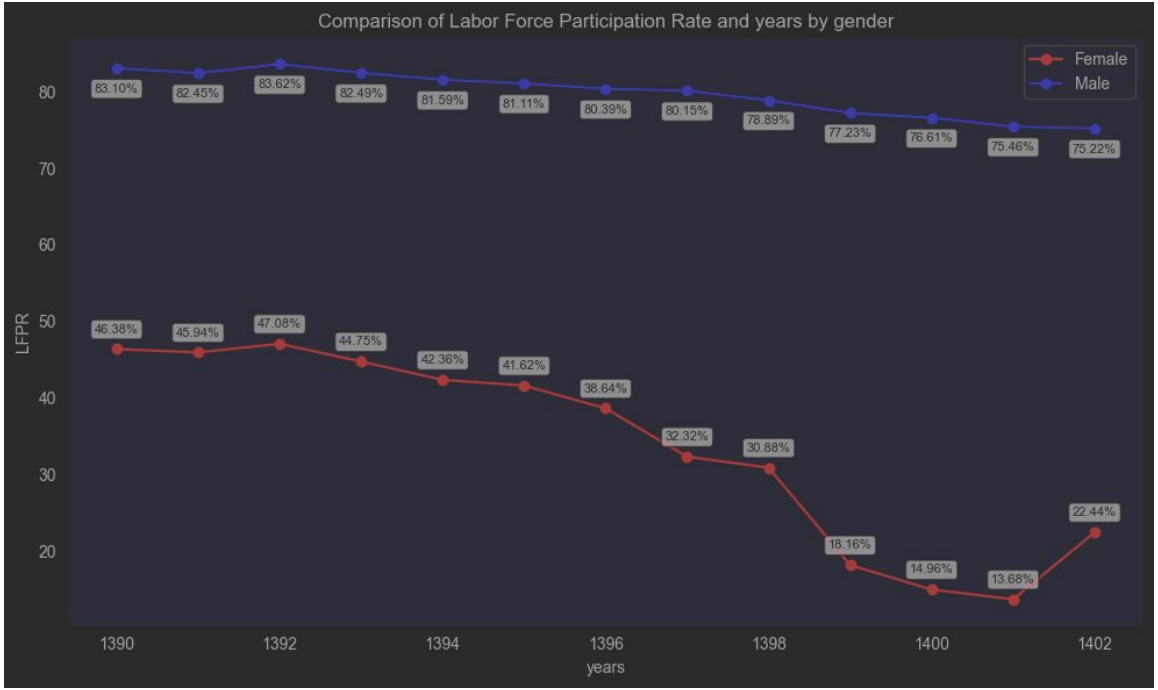
میانه و میانگین درآمد زنان و مردان



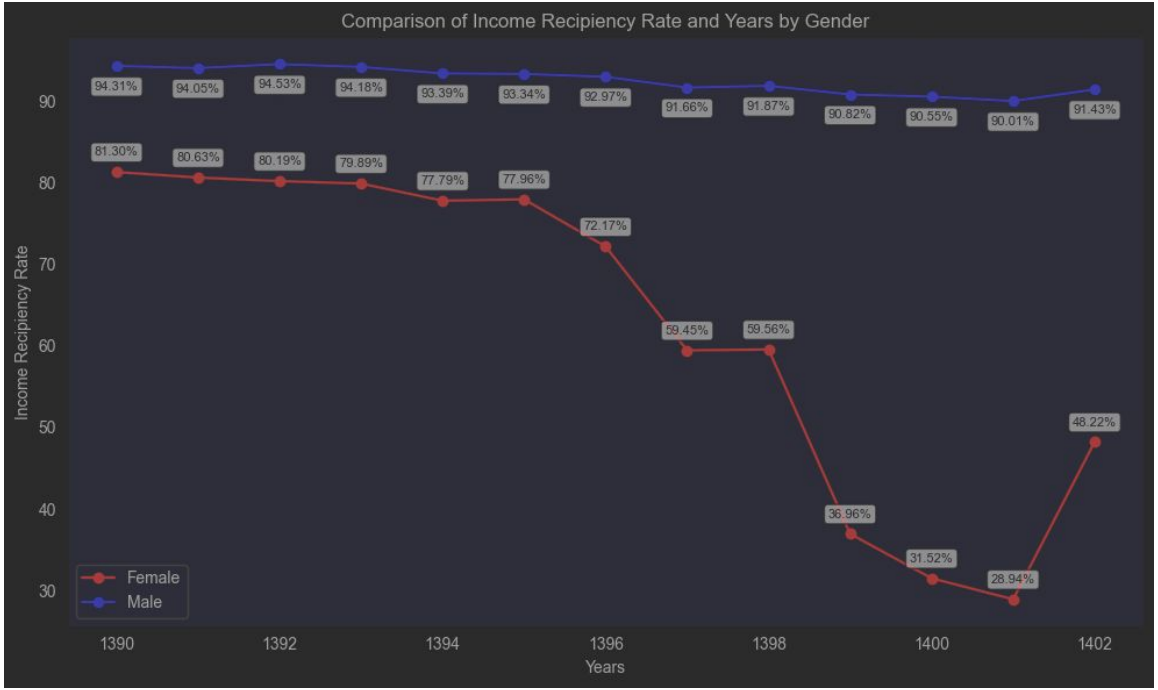
نرخ بیکاری


Comparison of Unemployment Rate and Years by Gender
Female
Male
Unemployment Rate
Years
7
6
5
4
3
2
1390
1392
1394
1396
1398
1400
1402
4.01%
4.02%
3.76%
4.12%
4.75%
4.98%
4.76%
5.10%
5.08%
5.78%
5.25%
5.54%
4.76%
1.60%
1.62%
1.54%
1.65%
1.97%
2.26%
2.65%
4.36%
4.26%
6.23%
6.38%
7.12%
6.27%

$$\text{LFPR} = \frac{\text{نیروی کار (شاغلان + بیکاران جویای کار)}}{\text{جمعیت در سن کار}} \times 100$$



$$\text{نرخ برخورداری از درآمد} = \frac{\text{تعداد افرادی که دارای درآمد هستند}}{\text{کل جمعیت مورد بررسی}} \times 100$$



جمع‌بندی: حلقه معیوب

- خونه اول: دیوار بلند عدم مشارکت (نرخ مشارکت نیروی کار پایین)
- خونه دوم: دریچه تنگ اشتغال (نسبت اشتغال بسیار پایین)
- خونه سوم: انسداد روزافزون بازار کار (صعود شدید نرخ بی‌کاری زنان)
- خونه چهارم: فروریزی آخرین پایگاه های درآمدی (سقوط آزاد نرخ برخورداری از درآمد)
- خونه پنجم: سقف شیشه‌ای درآمد (شکاف دستمزد و درآمد پایین)

پرسش و پاسخ